Calculate Error:

Actual output = 1

Predicted output = 0.75

Find the error

Sol: Error = Target − Output

$= 1 - 0.75$

$= 0.25$

Find: Mean Squared Error (MSE)

Given Targed outputs 1, 0, 1

Predicted outputs 0.8, 0.3, 0.9

$$MSE = \frac{1}{3}\left[(1-0.8)^2 + (0-0.3)^2 + (1-0.9)^2\right]$$

$$= \frac{1}{3}(0.04 + 0.09 + 0.01)$$

$$= \frac{0.14}{3}$$

$$MSE = 0.0467$$

5/6

Find new weight = Oldweight + n × Error × input

$= 0.5 + (0.2)(0.4)(1)$

$= 0.5 + 0.08$

$= 0.58$

Gradient Descent

Gradient Descent is an optimization algorithm that minimizes the error by updating the weights

2) Stochastic (SGD)

The model updates its weights after processing each individual training sample, rather than waiting for entire dataset.

Working of SGD

Sample	Target
x_1	10
x_2	15
x_3	20

Training Sequence:-

Input $x_1 \rightarrow$ updat weights

Input $x_2 \rightarrow$ update weights

Input $x_3 \rightarrow$ update weights

3) Mini-Batch gradient Descent

(MBGD) uses a small batch of samples

ex:- Dataset 1000 images

Batch size:- 100 images

weight updates:- 100 images $\rightarrow$ update

Next 100 images $\rightarrow$ update

Next 100 images $\rightarrow$ update

Number of hidden layers : 4

Model Capacity

Model Capacity refers to the ability of a neural network to learn to Patterns

Network Architecture

Types of Neural Network

a) Feed forward Neural Network (FNN)

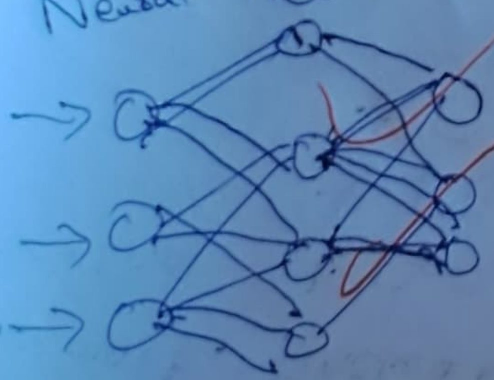
-> Information flows in one direction

-> No feed loops

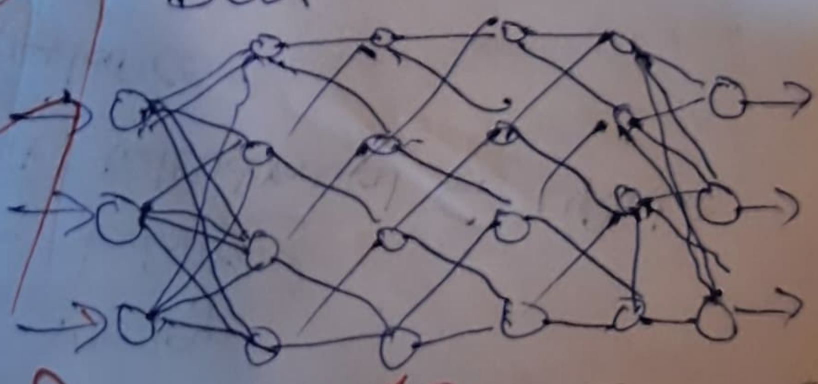
-> used for Classification

b) Convolutional Neural Network

Neural Networks



Deep Neural Networks



Binary Entropy

Sol: Probability of Pass:

$$P(\text{Passs}) = \frac{8}{10} = 0.80$$

Probability of fail:

$$P(\text{fail}) = \frac{2}{10} = 0.2$$

2. Multi-Class Entropy

Definition

Multi-Class entropy is used when there are more than two Classes

ex: an image classification problem may have:

* Cat $= P(\text{Cat}) = \frac{4}{12} = \frac{1}{3}$
* Dog $= P(D) = \frac{4}{12} = \frac{1}{3}$
* Horse $= P(H) = \frac{4}{12} = \frac{1}{3}$

1/8/2026